

VPA ICT Program — Grades 6–8 Course Overview

VPA ICT Program — Grades 6–8: Digital Mastery

Academic Years: 2024 – 2027 (3-year rotation) | Lesson Duration: 40 minutes
Target Group: EAL learners (Mandarin L1), mixed ability | Standards: CSTA 6–8, ISTE Standards

Program Vision

The Grades 6–8 ICT program prepares students for IGCSE ICT (0417) by developing advanced proficiency in productivity suites, introducing programming with Python, exploring web design fundamentals, and building awareness of cybersecurity and data analysis. Each year has a distinct focus while reinforcing prior learning.

Three-Year Rotation

Year	Focus	Key Topics	Standards
Year 6 (G6)	Advanced Productivity & Web Basics	Advanced Office, HTML/CSS intro, cloud tools	CSTA 2-DA-08, ISTE 1.6, 1.7
Year 7 (G7)	Programming & Databases	Python fundamentals, database concepts, data analysis	CSTA 2-AP-10, 2-DA-09, ISTE 1.5
Year 8 (G8)	Cybersecurity, Robotics & Media	Security, micro:bit, video editing, data visualization	CSTA 2-IC-22, 2-AP-12, ISTE 2.1

Skills Progression Map

Skill Area	Year 6	Year 7	Year 8
Word Processing	Advanced formatting, mail merge, styles	Review & extension	Review & IGCSE prep
Spreadsheets	Advanced formulas, conditional formatting	Functions, pivot tables, data validation	Data analysis, what-if scenarios
Presentations	Master slide, multimedia	Data-driven presentations	Professional pitching
Web Design	HTML tags, CSS basics	Multi-page sites, forms	Responsive design intro
Programming	Block-to-text transition	Python: variables, loops, if/else	Python: functions, lists, libraries
Databases	Introduction to records	Tables, queries, sorting	Relational databases, forms
Cybersecurity	Password security basics	Phishing, encryption concepts	Network security, ethical hacking
Robotics	—	—	micro:bit, sensors, Arduino intro
Media	Digital image editing	Audio editing	Video editing, green screen

Assessment Overview

Formative:

Weekly coding challenges (auto-graded on Replit/CodeHS)

Project milestones with peer review

Debugging journals

Vocabulary quizzes (bilingual)

Summative:

Year 6: Integrated Office project + 3-page website

Year 7: Python application + database project

Year 8: Security audit report + micro:bit invention + video project

IGCSE ICT Readiness Checklist

By the end of Grade 8, students should be able to:

- ☐ Create a multi-page document with styles, tables of contents, and mail merge
- ☐ Build a spreadsheet with complex formulas, charts, and pivot tables
- ☐ Design a 10+ slide presentation with master slides and embedded media
- ☐ Write a Python program using variables, loops, if/else, and functions
- ☐ Create a functional 3-page website with HTML and CSS
- ☐ Query a database using simple SQL SELECT, WHERE, ORDER BY
- ☐ Explain 5+ cybersecurity threats and protection methods
- ☐ Produce a 2-minute edited video with titles and transitions